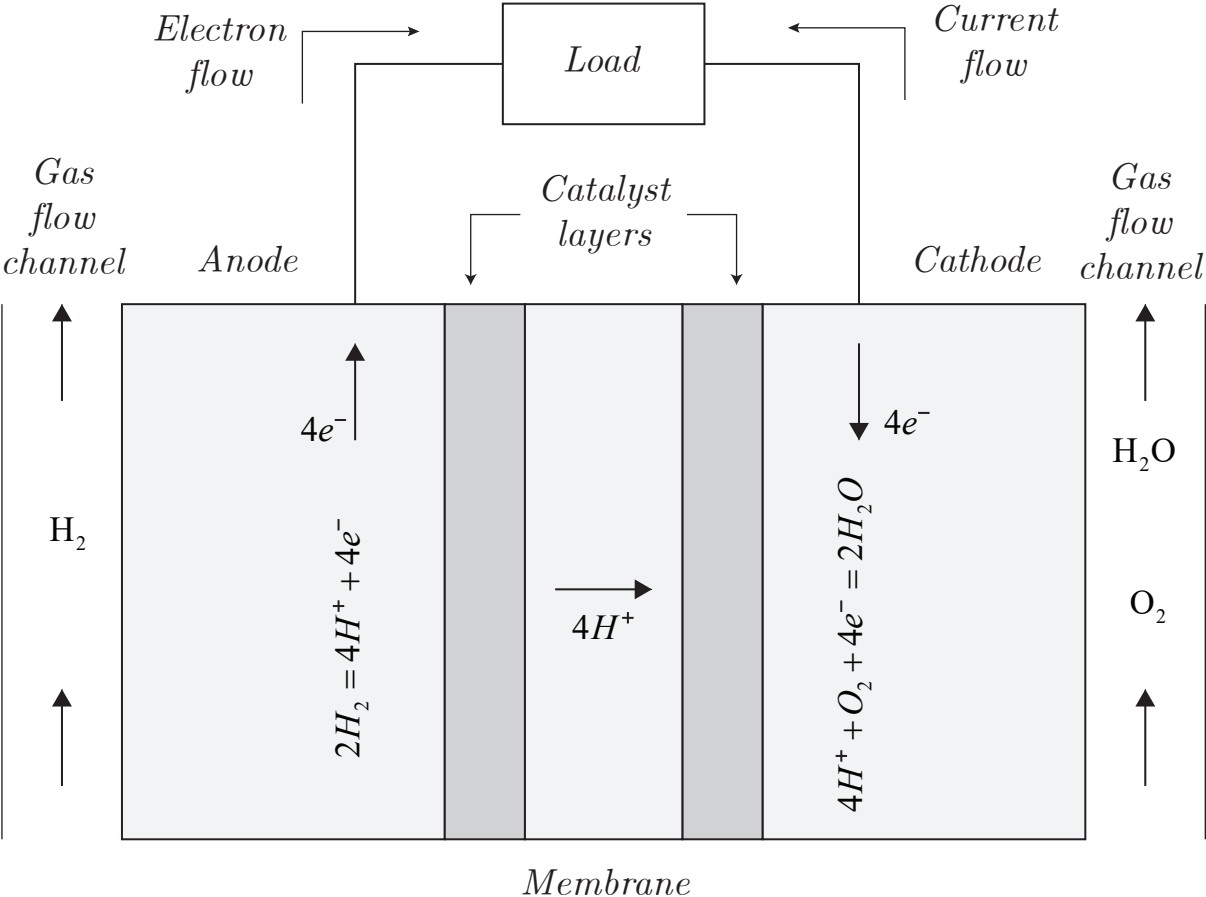
Electron flow
Load
Current flow
Gas flow channel
Anode
Catalyst layers
Cathode
Gas flow channel
$4e^-$
$4e^-$
H_2
H_2O
$2H_2 = 4H^+ + 4e^-$
$4H^+$
$4H^+ + O_2 + 4e^- = 2H_2O$
O_2
Membrane

$$2H_2 + O_2 = 2H_2O$$